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Case 2-2022: A 70-Year-Old Man with a Recurrent Left Pleural Effusion

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PRESENTATION OF CASE

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N Engl J Med 2022;386:274-83.
 DOI: 10.1056/NEJMcpc2115847
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Dr. Cory A. Perugino: A 70-year-old man was evaluated in the rheumatology clinic of this hospital because of a recurrent left pleural effusion.

Five years before this presentation, the patient had pain in the shoulders and proximal muscles of the arms and legs. The pain was worse in the morning and was associated with fatigue. There was no associated stiffness. The erythrocyte sedimentation rate and the C-reactive protein level were elevated. Prednisone was initiated for the treatment of a working diagnosis of polymyalgia rheumatica, and symptoms abated quickly. During the next year, attempts to discontinue prednisone led to recurrent myalgias, as well as elevations in the erythrocyte sedimentation rate and the C-reactive protein level.

Three years before this presentation, swelling, erythema, and pain developed in the cartilaginous portion of the left ear, sparing the earlobe (Fig. 1). There was no reduction in these symptoms after empirical antibacterial therapy was initiated for the treatment of presumed cellulitis. The dose of prednisone was increased, and the symptoms involving the ear resolved; however, they recurred when the dose of prednisone was tapered. Biopsy of the left auricle revealed perichondrial inflammation and degenerative changes in the cartilage. The dose of prednisone was increased for a presumed diagnosis of relapsing polychondritis. The symptoms involving the ear resolved, and weekly therapy with methotrexate was initiated.

During the next 6 months, attempts to taper the prednisone dose were associated with worsening inflammatory joint and cartilage symptoms, along with elevations in the erythrocyte sedimentation rate and the C-reactive protein level. The erythrocyte sedimentation rate was as high as 89 mm per hour (reference range, 0 to 13) and the C-reactive protein level as high as 73.6 mg per liter (reference value, <8.0).

Two years before this presentation, the patient began treatment with tocilizumab for fatigue, markedly elevated C-reactive protein level, and elevated erythrocyte sedimentation rate, all of which persisted despite treatment with prednisone and methotrexate. The administration of tocilizumab allowed for the tapering of

Figure 1. Clinical Photographs.

A photograph of the posterior portion of the left ear (Panel A) obtained 3 years before this presentation shows prominent auricular edema and erythema; a small portion of the ear (the noncartilaginous portion) is unaffected. These findings are consistent with chondritis. The unaffected right ear (Panel B) is shown as a reference to highlight the degree of inflammation involving the contralateral auricle at the time of onset of chondritis. A photograph of the left eye (Panel C) obtained 15 months before this presentation shows ocular injection (i.e., redness), which was more prominent in the left eye than in the right eye, with sparing of the limbus — a finding that is consistent with scleritis. This sign was associated with ocular pain and reduced visual acuity.



prednisone to a lower dose without causing an exacerbation of inflammatory joint and cartilage symptoms. Two weeks after the patient began receiving tocilizumab therapy, and upon tapering of the prednisone dose, pain and swelling developed in the right leg; ultrasound examination revealed a distal deep-vein thrombosis in the right posterior tibial vein. Repeat ultrasonography of the leg performed 2 weeks later did not show propagation of the deep-vein thrombosis, and the pain and swelling resolved with the continued use of a compression stocking and elevation of the right leg. However, 2 months later, when further tapering of the prednisone dose was started, inflammatory joint and cartilage symptoms recurred, and pain and swelling developed in the left leg. Ultrasound examination revealed deep-vein thromboses in the left popliteal and left calf veins and in the right femoral vein; therapy with rivaroxaban was initiated. The development of recurrent deep-vein thromboses was attributed to the underlying inflammatory disease. The dose of prednisone was increased, methotrexate was continued, and tocilizumab was replaced with infliximab therapy.

During the next 6 months, the inflammatory joint and cartilage symptoms worsened when attempts were made to taper the prednisone dose, and these conditions abated only when the dose of prednisone was increased. Blood tests for antineutrophil cytoplasmic antibodies (ANCA), rheumatoid factor, and anti-cyclic citrullinated peptide antibodies were negative. Fifteen months before this presentation, treatment with rituximab was initiated. However, after an attempt to taper the dose of prednisone, bilateral scleritis developed (Fig. 1); the dose of prednisone was

increased again, which resulted in resolution of the ocular pain and redness. During the next 3 months, attempts to taper the prednisone dose

resulted in recurrent episodes of scleritis and inflammatory joint and cartilage symptoms; additional rituximab treatment was not administered.

One year before the current presentation, treatment with low-dose cyclophosphamide was initiated, but the use of this medication was complicated by severe neutropenia, which did not allow for the dose to be escalated. Nine months before this presentation, treatment with cyclophosphamide was stopped, and intravenous tocilizumab was restarted, given its previous efficacy in this patient. However, the use of tocilizumab also resulted in moderate but transient neutropenia after each infusion.

Seven months before this presentation, pleuritic chest pain on the lower left side developed, along with dyspnea on exertion and a dry cough. Chest radiography revealed a moderate left pleural effusion (Fig. 2A and 2B). Thoracentesis was performed, and 1 liter of straw-colored pleural fluid was removed; laboratory test results are shown in Table 1. Gram's staining of the pleural fluid revealed abundant polymorphonuclear cells but no organisms; bacterial, fungal, and mycobacterial cultures were without growth. Cytologic examination of the pleural fluid revealed no malignant cells. Computed tomography (CT) of the chest (performed after the pleural fluid was drained) showed a small left pleural effusion, a trace right pleural effusion, and bibasilar dependent atelectasis without consolidation, nodules, or lymphadenopathy (Fig. 2C). The pleural effusions were attributed to ongoing inflammatory disease. The dose of prednisone was increased, and the pleurisy and dyspnea resolved. Five months before this presentation, tocilizumab therapy was stopped, and tofacitinib was initiated. One week before this presentation, the patient had sudden onset of back pain on the left side. Imaging studies were performed.

Dr. Vincent V. Dinulescu: CT of the chest and abdomen, performed 6 months after the first CT of the chest, revealed a small left pleural effusion that was larger than it had been on the first chest CT (Fig. 2D). A punctate stone in the lower pole of the left kidney was also observed, with no evidence of hydronephrosis.

Dr. Perugino: The patient presented to the rheumatology clinic for further evaluation. He had no dyspnea, cough, or fever. On examination, the temperature was 36.7°C, the heart rate 64 beats per minute, the blood pressure 145/75 mm Hg,

the respiratory rate 18 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 29.2. There was no scleral redness or erythema of the helix of the ear. The heart sounds were regular. Breath sounds were decreased at the base of the left lung. There was no leg edema, joint swelling or tenderness, purpura, petechiae, or deficits in motor strength.

The patient had a history of mucinous adenocarcinoma of the cecum, which had been diagnosed 10 years before this presentation and had been treated with laparoscopic colectomy of the right side of the colon and with oxaliplatin therapy, without evidence of recurrence. Other history included peripheral neuropathy (attributed to the oxaliplatin therapy), hypothyroidism, and monoclonal gammopathy of undetermined significance. Macrocytic anemia had been present for 3 years before this presentation, with normal blood levels of cobalamin and folate. The patient also had a history of leukopenia and neutropenia; these conditions had occurred intermittently over the course of the 10 years preceding this presentation but were most pronounced after treatment with cyclophosphamide and tocilizumab. Medications included alendronate, atorvastatin, cholecalciferol, folic acid, levothyroxine, methotrexate, omeprazole, prednisone, rivaroxaban, trimethoprim-sulfamethoxazole, and tofacitinib. The patient drank alcohol occasionally; he did not smoke tobacco or use illicit drugs. He lived in New England and was retired. There was no family history of autoimmune disease.

The blood levels of electrolytes, glucose, and thyrotropin were normal, as were the results of liver-function and kidney-function tests. Testing for antinuclear antibodies was negative. Other laboratory test results are shown in Table 1.

A diagnostic test was performed, and management decisions were made.

DIFFERENTIAL DIAGNOSIS

Dr. Peter C. Grayson: This 70-year-old man presented with a recurrent left pleural effusion in the context of a severe, systemic illness that was characterized by musculoskeletal pain, chondritis, recurrent deep-vein thrombosis, scleritis, and elevated levels of acute-phase reactants. His presentation was notable for a series of seemingly

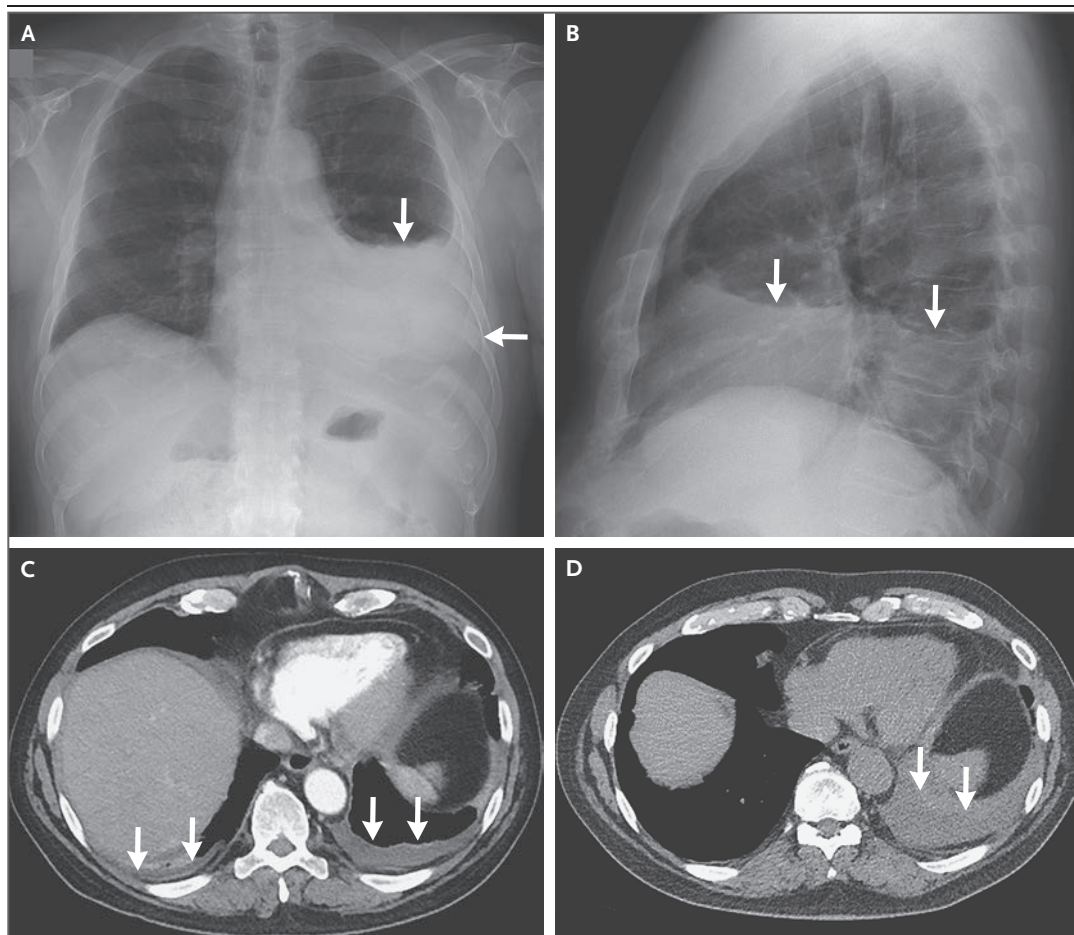


Figure 2. Imaging Studies of the Chest.

Posteroanterior (Panel A) and lateral (Panel B) chest radiographs obtained 7 months before this presentation show a moderate left pleural effusion (arrows). A contrast-enhanced axial CT image through the thorax (Panel C), obtained after thoracentesis, shows a small left pleural effusion and a trace right pleural effusion (arrows) with bibasilar atelectasis but no consolidation, nodules, or lymphadenopathy. Six months later, an axial CT image through the thorax (Panel D), obtained without intravenous contrast enhancement, shows a small left pleural effusion (arrows) that is larger than it was on the previous CT image.

unrelated and progressive inflammatory symptoms that were refractory to treatment with medications other than glucocorticoids.

POLYMYALGIA RHEUMATICA

This patient's initial symptoms of pain in the shoulders and hip girdle were diagnosed as polymyalgia rheumatica. Polymyalgia rheumatica is a misnomer, since this condition does not primarily involve muscle or cause weakness. Specifically, symptoms of pain or stiffness associated with polymyalgia rheumatica are caused by interspinous, subacromial, and trochanteric bursitis. The symptoms are extremely responsive to treatment

with low-to-moderate doses of glucocorticoids; however, gradual reduction of treatment over a period of many months is usually necessary before symptoms fully resolve. In suspected cases, it is important to maintain a high index of suspicion for concurrent diseases. In approximately 10 to 20% of patients who have polymyalgia rheumatica, giant-cell arteritis can develop.¹ This patient had no symptoms or vascular abnormalities that were suggestive of a concomitant large-vessel vasculitis. Polymyalgia rheumatica can be a harbinger of rheumatoid arthritis; however, this patient had no features of peripheral arthritis. Musculoskeletal symptoms within the

Table 1. Laboratory Data.

Variable	Reference Range, Adults*	7 Mo before Presentation	On Presentation
Blood			
Hemoglobin (g/dl)	13.7–17.5	12.4	11.1
Hematocrit (%)	41.0–53.0	37.1	33.7
Mean corpuscular volume (fl)	80–100	108.5	112.7
White-cell count (per μ l)	4500–11,000	5700	3030
Differential count (%)			
Neutrophils	40–70	80.4	49.6
Lymphocytes	22–44	7.7	32.3
Monocytes	4–11	6.5	12.5
Eosinophils	0–8	1.2	2
Basophils	0–3	0.7	1
Platelet count (per μ l)	150,000–400,000	165,000	246,000
Erythrocyte sedimentation rate (mm/hr)	0–13	55	107
C-reactive protein (mg/liter)	<8	88.7	205.8
Lactate dehydrogenase (U/liter)	110–210	192	—
Total protein (g/dl)	6.0–8.3	7.2	7.4
Pleural fluid			
Lactate dehydrogenase (U/liter)	—	195	—
Total protein (g/dl)	—	4.8	—
Albumin (g/dl)	—	3.3	—
Glucose (mg/dl)	—	111	—
Red-cell count (per μ l)	—	55,000	—
Total nucleated cells (per μ l)	—	2742	—
Differential count (%)			
Neutrophils	0	30	—
Lymphocytes	0	51	—
Macrophages	0	11	—
Monocytes	0	2	—
Eosinophils	0	3	—
Basophils	0	1	—
Unclassified cells	0	2	—

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

spectrum of polymyalgia rheumatica have been reported in association with various cancers, particularly hematologic neoplasms.

AURICULAR CHONDritis

This patient had clinical features of auricular chondritis, including cartilaginous inflammation in the external ear, sparing the noncarti-

laginous earlobe. Symptoms of chondritis are not commonly associated with polymyalgia rheumatica. The differential diagnosis for systemic diseases that are manifested by symptoms of auricular chondritis is limited. Relapsing polychondritis is an inflammatory disease characterized in part by cartilaginous inflammation of the ear, nose, airway, and joints.² Chondritis of

the nose and airway can be subtle and overlooked. It is important to obtain additional information from the patient, including whether he had nasal pain, difficulty breathing, cough, neck tenderness, or chest-wall pain. Given the morbidity and mortality associated with chondritis of the airway, the presence of symptoms involving the airway warrants further evaluation for subglottic stenosis and tracheobronchomalacia by direct visualization of the airway or non-invasive imaging.

Cartilaginous inflammation of the ear, nose, and airway is also a feature of granulomatosis with polyangiitis (GPA). The clinical features of GPA overlap with those of relapsing polychondritis. The absence of detectable ANCA and of kidney involvement in this patient is more consistent with relapsing polychondritis than with GPA. This patient subsequently had scleritis, which is a feature of both GPA and relapsing polychondritis. Although infections (including mycobacterial infections) also cause chondritis of the ear in rare instances, they are unlikely to be the cause of this patient's presentation because of the chronic, relapsing nature of the ear pain, with a reduction in symptoms noted in response to glucocorticoid therapy.

RECURRENT THROMBOEMBOLIC DISEASE

Another notable feature of this patient's history is the development of deep-vein thrombosis. Recurrent deep-vein thrombosis raises the possibility of an underlying hypercoagulable disease or occult cancer, especially given the history of gastrointestinal adenocarcinoma. Thromboembolic disease occurs in association with a broad spectrum of systemic inflammatory diseases, including many forms of vasculitis.³

EXUDATIVE PLEURAL EFFUSION

Seven months before this patient's current presentation, he had a left pleural effusion. Analysis of the pleural fluid revealed a sterile, exudative effusion without cytologic evidence of cancer. Exudative pleural effusions are associated with certain rheumatologic diseases. Serositis is common in patients with systemic lupus erythematosus; however, this diagnosis can essentially be ruled out in this case, given the patient's undetectable levels of antinuclear antibodies and absence of other supporting features. Pleural effusions in patients with rheumatoid arthritis are

characteristically associated with a low pH and low glucose level in pleural fluid⁴; we are not given the pH of this patient's pleural fluid, but the level of glucose was not low. Exudative effusions occur rarely in patients with GPA but are not a feature of relapsing polychondritis. It is important to consider autoinflammatory diseases, such as familial Mediterranean fever and tumor necrosis factor receptor–associated periodic syndrome, in cases of recurrent serositis with systemic inflammation and musculoskeletal pain. The development of an exudative pleural effusion later in the disease course expands, rather than narrows, the differential diagnosis in this case.

HEMATOLOGIC ABNORMALITIES

This patient had laboratory abnormalities that were consistent with macrocytic anemia, neutropenia, and a monoclonal gammopathy. Anemia of chronic disease, which is commonly associated with systemic inflammatory diseases, is characterized by normocytic or microcytic anemia. Macrocytic anemia is exceedingly uncommon in most rheumatologic diseases, unless it is caused by toxic effects of medication. After common causes of macrocytic anemia have been ruled out, this finding may signify a bone marrow disease such as a myelodysplastic syndrome. Concomitant myelodysplastic syndromes have been reported in association with a spectrum of inflammatory diseases, including relapsing polychondritis among older men.^{5,6} Toxic effects resulting from medication use are the most common cause of neutropenia in patients with rheumatologic diseases; however, neutropenia may be a feature of systemic lupus erythematosus or Felty's syndrome. Finally, monoclonal gammopathy suggests the possibility of multiple myeloma, a condition that can occur simultaneously with polymyalgia rheumatica.

VEXAS SYNDROME

In late 2020, a new disease entity that was referred to as VEXAS (vacuoles, E1 enzyme, X-linked, autoinflammatory, somatic) syndrome was reported.⁷ VEXAS syndrome is caused by somatic mutations in *UBA1* (the gene that encodes ubiquitin-like modifier activating enzyme 1), which initiates the first step of cellular ubiquitylation. The mutations arise later in life and are confined to hematopoietic precursor cells in bone

marrow and myeloid cells (e.g., neutrophils and monocytes) in peripheral blood. Since *UBA1* is an X-linked gene and women are protected by a second allele, VEXAS syndrome occurs almost exclusively in men. The majority of disease-defining mutations are missense mutations in codon 41 in *UBA1*, which impairs translation and function of the cytoplasmic isoform of the *UBA1* enzyme.

Patients with VEXAS syndrome can have inflammatory symptoms that mimic rheumatologic diseases.⁸ Specifically, patients with VEXAS syndrome can meet diagnostic or classification criteria for one or more diseases, including relapsing polychondritis, polyarteritis nodosa, giant-cell arteritis, Sweet's syndrome, ANCA-associated vasculitis, systemic lupus erythematosus, and adult-onset Still's disease.^{7,9,10} This patient had several clinical features that were consistent with VEXAS syndrome. Chondritis is a common clinical feature in patients with VEXAS syndrome, is typically confined to the ear or nose, and can be transient and subtle without resultant damage.⁷ Fever and cutaneous manifestations that reflect neutrophilic dermatosis, leukocytoclastic vasculitis, or medium-vessel vasculitis are common in patients with VEXAS syndrome but were not present in this case.⁷ Pulmonary involvement can occur, often in the later stages of disease. Sterile neutrophilic alveolitis is the most common pulmonary manifestation, but exudative pleural effusions have been reported in a minority of patients. Musculoskeletal conditions that can mimic polymyalgia rheumatica have been described. Recurrent thromboembolic disease, in the absence of traditional risk factors for hypercoagulability, is common in patients with VEXAS syndrome. Inflammation can be severe and difficult to treat, and only glucocorticoids have shown consistent effectiveness in ameliorating symptoms.^{7,11}

In addition to systemic inflammatory disease, progressive bone marrow failure is a defining feature of VEXAS syndrome. Mutations in *UBA1* in bone marrow stem cells and in hematopoietic progenitor cells, with subsequent clonal expansion, result in hematologic abnormalities. Macrocytosis is the most common hematologic manifestation and is typically an early feature of disease. Progressive cytopenias, especially thrombocytopenia and lymphopenia, occur in the later phases of disease.¹² Bone marrow biopsies that are performed in patients with VEXAS syndrome

typically reveal hypercellular marrow with myeloid predominance. Numerous cytoplasmic vacuoles in erythroid and myeloid precursor cells are present in the bone marrow aspirate in almost all patients with VEXAS syndrome.⁸ Monoclonal gammopathy has been reported in approximately 20% of patients.⁷ Serial bone marrow biopsies may reveal progression to myelodysplastic syndrome or to multiple myeloma.¹²

A simple clinical algorithm to identify patients with VEXAS syndrome with near-perfect accuracy has been proposed.¹³ Men with chondritis of the ear or nose are likely to have VEXAS syndrome if the mean corpuscular volume is greater than 100 fl or if the platelet count is greater than 200,000 per microliter. When the algorithm is applied to this patient, the findings suggest that genetic testing will confirm a diagnosis of VEXAS syndrome. A bone marrow biopsy should be performed to rule out hematologic cancer; however, genetic testing for VEXAS syndrome may be performed on either bone marrow or peripheral-blood cells, since mutational burden is typically high in both compartments.

CLINICAL IMPRESSION

Dr. Perugino: The episodic nature of the syndrome and responsiveness to glucocorticoids were suggestive of an autoimmune or autoinflammatory disease process, and the concurrence of inflammatory musculoskeletal pain, recurrent chondritis, scleritis, and pleural effusion supported the idea that all the features were linked by a single process. The most salient aspect of this patient's presentation was the profound refractoriness to multiple immunosuppressive agents. Paired with the smoldering and progressing macrocytosis, the primary diagnostic consideration was an autoinflammatory process caused by underlying myelodysplastic syndrome. A bone marrow biopsy was scheduled to evaluate for myelodysplastic syndrome. On the basis of the observed degree of refractoriness to multiple immunosuppressive medications and the presence of chondritis, macrocytosis, and deep-vein thromboses, VEXAS syndrome was our leading diagnosis.

CLINICAL DIAGNOSIS

VEXAS (vacuoles, E1 enzyme, X-linked, autoinflammatory, somatic) syndrome.

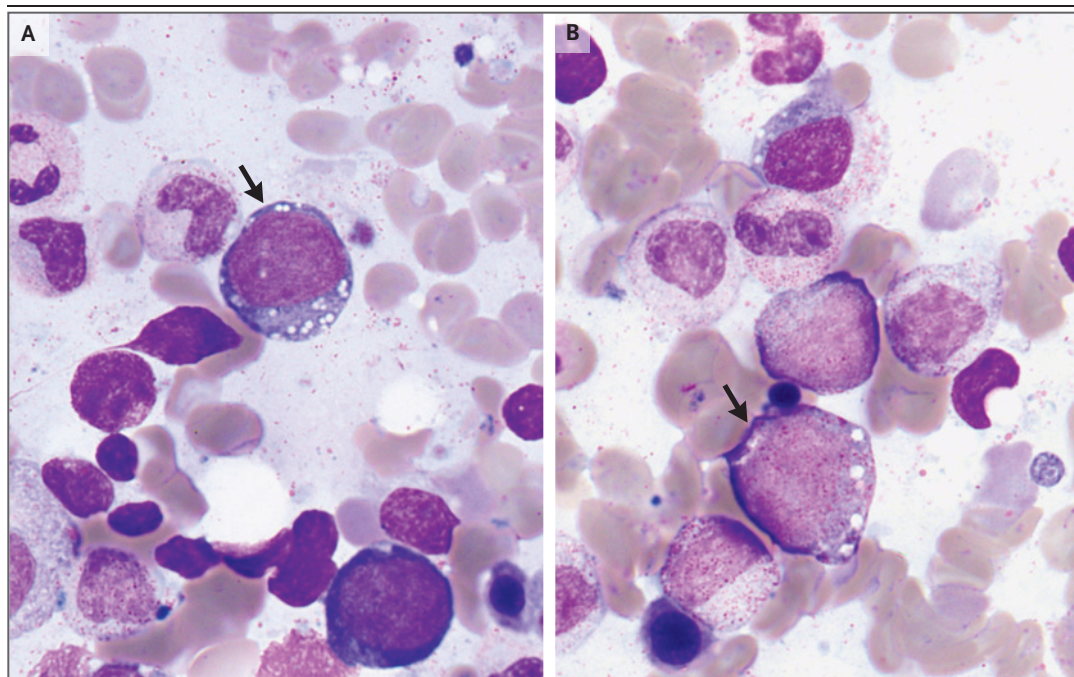


Figure 3. Bone Marrow Aspirate.

Wright–Giemsa staining of aspirate shows multiple vacuoles in a myeloblast (Panel A, arrow) and in a promyelocyte (Panel B, arrow).

DR. PETER C. GRAYSON'S DIAGNOSIS

VEXAS (vacuoles, E1 enzyme, X-linked, autoinflammatory, somatic) syndrome.

DIAGNOSTIC TESTING

Dr. Judith A. Ferry: The diagnostic procedure was a bone marrow biopsy. The core-biopsy specimen was hypercellular for the patient's age. The marrow specimen showed maturing trilineage hematopoiesis with an increased myeloid-to-erythroid ratio. The aspirate smear showed complete maturation of myeloid and erythroid elements. Multiple cytoplasmic vacuoles were present, mainly in early myeloid elements (including promyelocytes, occasional myelocytes, and rare blasts) and infrequently in early erythroid precursors (Fig. 3). Vacuoles were not conspicuous in more mature myeloid or erythroid elements or in lymphoid cells. On iron staining, an aspirate smear showed decreased iron stores and no ring sideroblasts.

Flow cytometry revealed 5% T cells (with no immunophenotypic abnormalities and a normal ratio of CD4 cells to CD8 cells), less than 1%

natural killer cells, no detectable B cells, and no increase in myeloid blasts. Next-generation sequencing revealed single-nucleotide variants in *DNMT3A* and *TET2*; in the absence of morphologic evidence of dysplasia, these variants were interpreted as clonal hematopoiesis of indeterminate potential. Cytogenetic analysis showed a normal male karyotype (46,XY).

In addition, peripheral blood was submitted to the National Human Genome Research Institute of the National Institutes of Health. A single variant (c.122T→G; p.Met41Thr) was identified in the *UBA1* gene by means of Sanger sequencing. In conjunction with the clinical and pathological findings in this case, this genetic abnormality confirms a diagnosis of VEXAS syndrome. The p.Met41Thr variant is the most common mutation that causes VEXAS syndrome.^{7,8} The presence of cytoplasmic vacuoles, as seen in this case, is not a specific finding; they can also be seen in marrow elements in association with zinc toxicity, copper deficiency, alcohol use, and myelodysplastic syndrome, as well as in certain types of neoplastic cells. However, the mutation in *UBA1* indicates that the vacuoles are related

to VEXAS syndrome and not to some other cause.

PATHOLOGICAL DIAGNOSIS

VEXAS (vacuoles, E1 enzyme, X-linked, autoinflammatory, somatic) syndrome.

DISCUSSION OF MANAGEMENT

Dr. Perugino: Glucocorticoids, with gradually escalating dose requirements over time, were the only consistently effective intervention to control this patient's acute inflammatory episodes. Early in what we now know to be the disease course of VEXAS syndrome in this patient, low doses of prednisone, typically less than or equal to 10 mg daily, were sufficient to prevent exacerbation of disease. However, by 3 to 4 years after the onset of disease, higher doses — typically in the range of 20 to 30 mg daily — were needed to prevent flares. Among the glucocorticoid-sparing agents that were used in this patient, interleukin-6 receptor antagonism with intravenous tocilizumab allowed for the lowest daily dose of prednisone, but the treatment was complicated by severe neutropenia. Empirical treatment with rituximab did not provide a glucocorticoid-sparing effect and resulted in persistent depletion of CD19+ B cells in the peripheral blood for nearly 30 months. After making a definitive diagnosis of VEXAS syndrome, we have considered allogeneic stem-cell transplantation, since mutations in *UBA1* are restricted to hematopoietic cells and are causal for the clinical phenotype of VEXAS syndrome. At this time, overt myelodysplastic syndrome has not yet been identified on bone marrow analysis and transfusion-dependent cytopenias have not yet developed; the presence of either of these would offset the risk of stem-cell transplantation.

PATIENT PERSPECTIVE

The Patient: Because VEXAS syndrome was not identified until the end of 2020, the past 5 years have been an ongoing time of trial and error around treatments to alleviate the symptoms of my then-unidentified disease. How frustrating it was for me to hear, over and over again, that we needed to switch to a new drug because the current one was just not working.

After my disease was diagnosed as VEXAS syndrome, I was relieved to know exactly what disease I had been dealing with all this time. But even now, I am unsure what my treatment will need to be in the future. Stem-cell transplantation is not high on my “bucket list,” but if it does become necessary, I would consider accepting the risks rather than continuing to have my life affected by VEXAS syndrome. I am hopeful that doctors will eventually find a treatment that will control this disease once and for all.

FUTURE DIRECTIONS

Dr. Grayson: The discovery of VEXAS syndrome has important clinical implications. The disease was identified with the use of a genotype-first approach, whereby a large genetic database was queried for novel genetic variants agnostic to clinical phenotype to identify a set of patients with somatic mutations in *UBA1*.⁷ As databases of genetic-sequencing data continue to accumulate, discovery of new diseases that may be difficult to recognize at the bedside is possible. VEXAS syndrome also highlights the concept that somatic mutations are causal in adult-onset inflammatory diseases.⁷ Somatic mutations occur after conception, are tissue-restricted, accumulate with age, and are not heritable if they occur outside of gonadal tissue. As such, somatic mutations will probably play a causal role in a spectrum of diseases other than cancer, especially in conditions in which onset occurs at advanced ages.¹⁴ Finally, VEXAS syndrome represents a new class of hematologic and inflammatory diseases that is defined by somatic mutations in hematopoietic cells with resultant systemic inflammation and progressive pathologic effects on the bone marrow.⁸

This case, and VEXAS syndrome in general, illustrates an interesting paradox in medical reasoning. In most systemic diseases, the differential diagnosis starts with a broad number of possible diagnoses and typically narrows over time, as new symptoms emerge, to ultimately converge on a final diagnosis. In VEXAS syndrome, the differential diagnosis starts with a broad list but continues to widen over time, as new disease features lead to additional diagnostic possibilities. Recognizing this pattern can be important to “converging” on the correct diagnosis.

This case was presented at the Harvard Medical School 2021 postgraduate course “Advances in Rheumatology,” organized by John H. Stone, M.D., M.P.H.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Lorena Wilson, N.P., Marcela Ferrada, M.D., David Beck, M.D., Ph.D., and Daniel Kastner, M.D., Ph.D. (all from the National Institutes of Health), for the coordination and performance of genetic testing for VEXAS syndrome in this patient.

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